E1. Rudolf and Snowflakes (simple version)

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: brute force, implementation, math | https://codeforces.com/problemset/problem/1846/E1

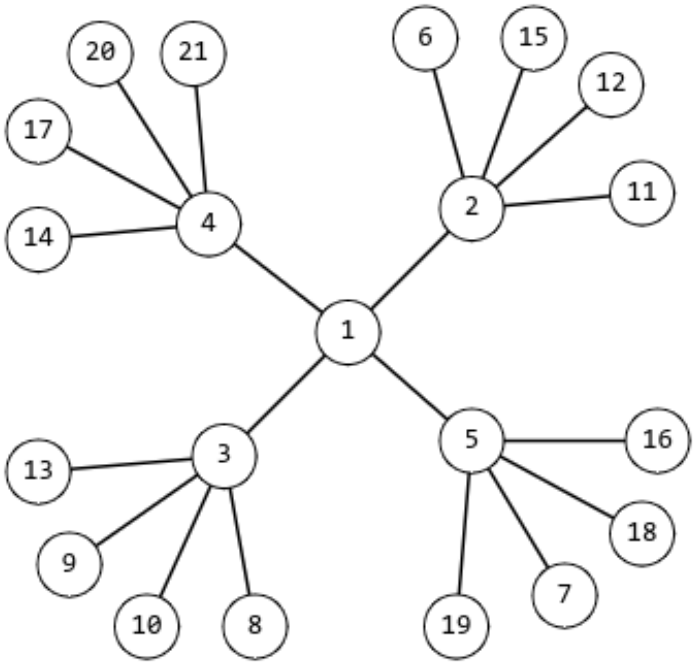
This is a simple version of the problem. The only difference is that in this version $n \le 10^6$.

One winter morning, Rudolf was looking thoughtfully out the window, watching the falling snowflakes. He quickly noticed a certain symmetry in the configuration of the snowflakes. And like a true mathematician, Rudolf came up with a mathematical model of a snowflake.

He defined a snowflake as an undirected graph constructed according to the following rules:

- Initially, the graph has only one vertex.
- Then, more vertices are added to the graph. The initial vertex is connected by edges to k new vertices (k gt 1).
- Each vertex that is connected to only one other vertex is connected by edges to k more new vertices. This step should be done **at least once**.

The smallest possible snowflake for $k = 4$ is shown in the figure.



After some mathematical research, Rudolf realized that such snowflakes may not have any number of vertices. Help Rudolf check if a snowflake with n vertices can exist.

Input

The first line of the input contains an integer t ($1 \le t \le 10^4$) — the number of test cases.

Then follow the descriptions of the test cases.

The first line of each test case contains an integer n ($1 \le n \le 10^6$) — the number of vertices for which it is necessary to check the existence of a snowflake.

Output

Output t lines, each of which is the answer to the corresponding test case — "YES" if there exists such k gt 1 for which a snowflake with the given number of vertices can be constructed; "NO" otherwise.

Example

Input
9 1 2 3 6 13 15 255 10101 1000000

Output
NO NO NO NO YES YES YES YES NO